

Serena R. Scholz

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Education

Ph.D. in Earth and Planetary Sciences, Yale University (2022-present), expected defense Spring 2027

- Committee: Juan M. Lora*, Alexey V. Fedorov†, Mary-Louise Timmermans, John S. Wettlaufer
*Major Discourse Advisor, †Minor Discourse Advisor
- Student in Atmosphere, Ocean and Climate Dynamics. Coursework: Geophysical Fluid Dynamics, Climate Dynamics, Physical Oceanography, Earth & Env. Physics, Planetary Atmospheres.

M.S. in Earth and Environmental Sciences, University of Michigan (2020)

- Thesis: Use of high-resolution oxygen isotope sclerochronology to reconstruct seasonal-scale precipitation regimes; advised by Sierra V. Petersen.

B.S. in Earth and Environmental Sciences, University of Michigan (2019)

Professional Research Experience

2020-2022. Associate Research Analyst, CNA Institute for Public Research.

2019-2021. Research Assistant, Ocean & Climate Physics Group, Lamont-Doherty Earth Observatory.

2016-2020. Research Assistant, SCIPP Group, University of Michigan Stable Isotope Laboratory.

Peer-Reviewed Publications

⁺in prep. *in press

⁺**Scholz, S.R.** and J.M. Lora. The dynamic lifecycle of atmospheric rivers: insights from idealized models. In preparation for *Science Advances*.

⁺**Scholz, S.R.** and A.V. Fedorov. Contrasting Responses of the Tropical Indian and Pacific Oceans to an Anomalous Surface Heat Flux. In preparation for *Geophysical Research Letters*.

***Scholz, S.R.** and J.M. Lora. “Quick study: Rivers in the sky.” in press, *Physics Today*.

Scholz, S.R. and J.M. Lora (2025). Widespread increase in atmospheric river frequency and impacts over the 20th century. *AGU Advances* v. 6, e2025AV001888. doi:10.1029/2025AV001888

Scholz, S.R. and J.M. Lora (2024). Atmospheric rivers cause warm winters and extreme heat events. *Nature* v. 636, p. 640–646. doi:10.1038/s41586-024-08238-7.

Scholz, S.R., S.V. Petersen and B.M. Anderson (2024). Modern reconstructions of mean and seasonal-scale climate from coastal marine gastropods (Turritellidae). *Palaeogeography, Palaeoclimatology, Palaeoecology* v. 655, 112553. doi:10.1016/j.palaeo.2024.112553.

Scholz, S.R., R. Seager, M. Ting, Y. Kushnir, J.E. Smerdon, B.I. Cook, E.R. Cook, and S.H. Baek (2022), Changing hydroclimate dynamics and the 19th to 20th century wetting trend in the English Channel region of northwest Europe. *Climate Dynamics* v. 58, p. 1539–1553. doi:10.1007/s00382-021-05977-5.

O'Hora, H. E., Petersen, S. V., Vellekoop, J., Jones, M. M., and **Scholz, S. R.** (2022), Clumped-isotope-derived climate trends leading up to the end-Cretaceous mass extinction in northwestern Europe. *Climate of the Past*, v. 18, p.1963–1982. doi:10.5194/cp-18-1963-2022.

Baek, S.H., J.E. Smerdon, G.-C. Dobrin, J.G. Naimark, E.R. Cook, B.I. Cook, R. Seager, M.A. Cane, and **S.R. Scholz** (2020), A quantitative hydroclimatic context for the European Great Famine of 1315-1317. *Communications Earth & Environment*, v. 1 (1), no. 1, doi:10.1038/s43247-020-00016-3.

Scholz, S.R., S.V. Petersen, J.H. Escobar, C. Jaramillo, A. Hendy, W.D. Allmon, J.H. Curtis, B.M.

Anderson, N. Hoyos, J.C. Restrepo, and N. Perez (2020), Isotope sclerochronology indicates enhanced seasonal precipitation in northern South America (Colombia) during the Mid Miocene Climatic Optimum. *Geology* v. 48 (7), p. 668-672. doi: 10.1130/G47235.1.

Invited Talks

Princeton Climate Dynamics & Impacts Seminar. “Atmospheric rivers cause warm winters and extreme heat events.” March 2025.

Conference Presentations

* talk, † poster, ° virtual due to COVID-19

***Scholz, S.R.** and J.M. Lora. Atmospheric rivers and energy transport in a hierarchy of idealized models. *EGU26 General Assembly*, Vienna, Austria, May 2026.

***Scholz, S.R.** and J.M. Lora. Atmospheric Rivers in a Hierarchy of Idealized Models. *AMS Annual Meeting, Conference on Climate Variability and Change*, Houston, TX, Jan. 2026.

***Scholz, S.R.** and J.M. Lora. Atmospheric River Structure and Energy Transport in a Hierarchy of Idealized Models. *AGU Fall Meeting*, New Orleans, LA, Dec. 2025.

***Scholz, S.R.** and J.M. Lora. Global trends in atmospheric river temperatures, snowfall, and precipitation. *AGU Fall Meeting*, Washington, D.C., Dec. 2024.

***Scholz, S.R.** and J.M. Lora. Global impacts of atmospheric rivers on surface temperatures and heat fluxes. *International Atmospheric Rivers Conference*, La Jolla, CA, Jun. 2024.

***Scholz, S.R.** and J.M. Lora. Atmospheric rivers cause extreme heat excursions and anomalously warm temperatures. *AGU Fall Meeting*, San Francisco, CA, Dec. 2023.

†**Scholz, S.R.** and J.M. Lora. How hot are atmospheric rivers? *Yale Climate Day*, New Haven, CT, May 2023.

°**Scholz, S.R.** and S.V. Petersen. Challenges and advances in reconstructing mean and seasonal climate from turritellid gastropods. *6th International Sclerochronology Conference*, Sep. 2022.

°**Scholz, S.R.** and S.V. Petersen. Reconstructing seasonal-scale precipitation patterns from high-resolution oxygen isotope sclerochronology of coastal turritellid gastropods. *GSA 2020 Connects*, Oct. 2020.

†**Scholz, S.R.**, R. Seager, M. Ting, and Y. Kushnir. Centennial Timescale Variability in Euro-Mediterranean Hydroclimate Since the Little Ice Age. *AGU Fall Meeting*, San Francisco, CA, Dec. 2019.

†**Scholz, S.R.**, S.V. Petersen, et al.. Clumped isotope calibration for marine gastropods. *Michigan Geophysical Union*, Ann Arbor, MI, Apr. 2019.

***Scholz, S.R.**, S.V. Petersen, et al.. Isotope Sclerochronology and Tropical Seasonality During the Mid Miocene Climatic Optimum. *Goldschmidt*, Boston, MA, Aug. 2018.

†**Scholz, S.R.**, S.V. Petersen, J.H. Escobar. Miocene seashells record ancient coastal environments. *Michigan Geophysical Union*, Ann Arbor, MI, Apr. 2018.

Teaching Fellowships

- 2026 EPS 1400 Atmosphere, Ocean and Climate Change
- 2025 EPS 140 Atmosphere, Ocean, and Climate Change
- 2023 EPS 101 Climate Change
- 2022 EPS 110 Dynamic Earth

Service

- Chair, “Atmospheric Rivers: Processes, Impacts, Observations, and Uncertainties Oral II.” *AGU Fall Meeting 2025*.

- Reviewer: *Geophysical Research Letters, Journal of Geophysical Research: Atmospheres, Nature Communications, Communications Earth & Environment, Journal of Climate, Weather.*

Academic Awards and Fellowships

- Excellence in Teaching Prize, for EPS 1400 (2026)
- Hammer Prize, outstanding graduate student in Yale EPS (2025)
- Best Poster Presentation, Michigan Geophysical Union (2019)
- Earth and Environmental Sciences Dept. Undergraduate Academic Excellence Award (2019)
- Eugene and Elizabeth Singer Award for Academic Excellence in Geology (2018)
- Matt Kelley/Granader Family Prize for Excellence in Writing (2017)
- Leonard and Eileen Newman Prize for Writing (2017)
- Roy W. Cowden Memorial Fellowship for Writing (2017)